Response to Reviewers of *Characterization of coherent flow structures in brain ventricles* by Halvor Herlyng and Shawn C. Shadden, submitted to Journal of Biomechanics

Below, *reviewer comments are italicized,* followed by nonitalicized author response.

Reviewer #1
Major Issues

The FTLE fields in Figure 7 A seem to show qualitatively different patterns toward the bottom right part of the domain at t0=0.4 s for T=1.0 s and at t0=0.2 s for T=-1.0s. I therefore wonder if there is some irregular or perhaps erroneous velocity that occurs in that vicinity of space and time. I request that the authors please check this and determine why the FTLE fields look qualitatively different for these release times.

Comment: We thank the reviewer for the high attention to details. Firstly, as the Reviewer remarked later in the review (under "*Minor Issues*"), we performed a convergence study to verify solutions to the Navier-Stokes equations.
Our understanding of why the structures look qualitatively different for these release times is due to the flow reversals that occur in the ventricles over the times in question. These reversals are evident in the flow rate graphs in Figure 3: the flow rates in the aqueduct and the interventricular foramina are negative at the beginning of the cardiac cycle (negative is here defined as CSF flowing from inferior to superior parts of the ventricular system), become positive around 0.05 sec, and turn negative again at around 0.4 sec. Because of these reversals, the structures in the FTLE fields initially emerge from flow in one direction, before being pulled back (downward in the Figure) as flow reverses inferiorly (which is the case at 0.2 and 0.4s).

Action: We have modified Sec. 4.1 of the Discussion and Sec. 3.5 of the Results to discuss this inferior shift of the structures in the FTLE fields during flow reversal.

The Discussion would benefit from more detailed consideration of mixing properties of repelling LCSs (i.e., forward FTLE ridges) versus attracting LCSs (i.e., backward FTLE ridges). As I'm sure the authors know, repelling LCSs are the regions of strongest stretching of fluid elements (in forward time), while attracting LCSs are the regions of tracer accumulation that generate observable mixing fronts (if a tracer were in the flow, for example). The first paragraph of the Discussion says "...this vortex ring... contribut[es] to mixing and stirring" and also that the "vortex ring also traps fluid" which may appear contradictory and could be explained much more thoroughly. Overall, a more detailed discussion of the specific implications of the forward vs backward FTLE *ridges would be beneficial (definitely in the Discussion section, but perhaps elsewhere as well).*

Comment/Actions: We thank the Reviewer for the constructive comment. In response, we have edited the Discussion section to be more specific about the implications of the attracting and repelling LCSs (backward- and forward-time FTLE ridges) in defining the vortex boundary as well as stirring that results from their entanglement.

Minor Issues

I wrote a long comment as a major issue asking about whether the simulations are adequately converged, but then I saw the results in the Appendix. Please add a brief discussion of convergence with citations to the Appendix in the main text. This would be most useful around lines 409 and 443, so the reader is reassured that the simulations are well-converged.

Comment/Actions: We appreciate the feedback from the Reviewer. To ensure readers that the simulations are adequately converged, we have added a reference to the Appendix and the convergence study therein in the Methods sections 2.8.1 and 2.8.2, as suggested by the Reviewer.

Line 78: "multiple scales" -- perhaps specify in time and/or in space

Action: We have specified that we here refer to time and length scales.

Line 572: I recommend rephrasing "...FTLE fields computed with varying release times t0 show how the FTLE fields vary in time (Figure 7)." Since FTLEs are computed over a finite time window, it is a bit imprecise to interpret Figure 7 as their evolution over time. Writing "...FTLE fields vary for different temporal windows" would be more accurate.

Comment/Action: We thank the Reviewer for pointing this out. We typically think of the FTLE as a function of space (i.e. a field) and time for unsteady flows. To view the temporal evolution of the FTLE field, the release time \$t_0\$ is varied. We typically view the integration time \$T\$ as a parameter. This is perhaps written more cleanly as \$\sigma(\mathbf{x},t;T)\$ where the semicolon separates “independent variables” from parameters. In the paper, we chose to use \$t_0\$ instead of \$t\$ to make it more explicit that, procedurally, we vary the release time to understand how FTLE “varies in time”. To address this confusion, we have changed the sentence in question to read

“...how the FTLE fields vary with release time” instead of “in time”, which we agree could be a little ambiguous.

We note that varying either the release time or integration time would result in a “different temporal window” so we chose not to use this terminology here.

Typos/Very minor issues

line 102: delete "of"
Figure 1 caption, (B): delete extra period
Line 163: rephrase because "Especially E..." is currently a sentence fragment
Line 497: "peak-to-through" should be "peak-to-trough"
Line 564: delete "and"

Action: We have addressed all the minor issues pointed out by the Reviewer.

Reviewer #2
Summary of the evaluation:
This is, as far as I know, the first application of formal FTLE/LCS study on cerebral ventricular CSF, differentiating itself from the predominantly Eulerian flow analyses that dominate the field and going beyond the CSF exchange study of Yoshida et al. (2022, J Roy Soc Interface), which quantified mixing in the lateral ventricles but did not extract Lagrangian coherent structures. The technical implementation is carefully done. The general observation that ventricular CSF flow features organising Lagrangian structures dominated by tissue deformation (rather than by CSF production or motile cilia) is not surprising, but it has not been studied with the same technical rigor before.

While I do not have any major concerns regarding the computational framework itself, I do have reservations about the biological interpretations of the results. These stem from the choice and implementation of the driving forces in both models, which omit physiologically important time-dependent components: respiratory modulation in the human model, and muscle-contraction-driven transient flows in the zebrafish model.

Comment: We thank the Reviewer for acknowledging the technical implementation and rigor of our study. We also thank the reviewer for their extremely thorough review and insightful comments. Broadly, we acknowledge that the biological interpretations of the results may have been overstated, or at least not carefully qualified. More detailed comments and information about our actions follow in the responses below.

Major points:
1. The zebrafish model assumes steady Stokes flow driven by motile ependymal cilia in a static geometry, which is at odds with experimental data: spontaneous muscle contractions produce substantial transient CSF flows in the central canal (Thouvenin et al. 2020, cited in the manuscript), and brain ventricles (Olstad et al. 2019, also cited). This is particularly well illustrated in the supplementary video S7 of the Olstad paper, 7 seconds into the movie. Furthermore, while ciliary action appears to be more dominant than the cardiac driving force, the latter does not appear to be negligible. In particular the spontaneous muscle contractions are a serious concern for the LCS analysis. The structures identified in Fig. 10 require integration times of 34, 68 and 101 s, over which one more contraction can occur. In Section 3.6, the authors show for the human model that neglecting fluid inertia in the unsteady dynamics substantially changes the resolved Lagrangian structures, showing that the FTLE structures are sensitive to the time-dependent details of the flow. By the same reasoning, the transient muscle contractions omitted from the zebrafish model could reorganize its FTLE structures, so their omission is a comparable concern.

The match between the LCS reported in the manuscript and the compartmentalization seen in Olstad et al. reflects a snapshot of in vivo biology, so both the experiment and the model characterize the same cilia-dominated regime. However, the manuscript extrapolates beyond this regime to what LCS imply about CSF transport, mixing and developmental function in the embryo more generally. Such claims must be dropped or qualified, since the in vivo flow includes cardiac and contraction-driven transient components that the model omits. Alternatively (or in addition - this could be very valuable), they could perform a sensitivity analysis in which a time-dependent perturbation representative of muscle contraction and cardiac action is superimposed on the steady cilia-driven flow, with LCS recomputed.

Comment: We thank the reviewer for raising these points. We agree that some of the conclusions were overly broad and should be refined, removed or qualified as appropriate. It is true that omitting cardiac driving forces and muscle contractions from our zebrafish model makes our model represent only a snapshot of in vivo flow. We would however like to note that:

1. Whilst working on the manuscript, we also computed FTLE fields and characterized LCS with a zebrafish model that included both motile cilia forcing and cardiac driving forces in the CSF model (this model was used in Herlyng, Halvor, et al. "Advection versus diffusion in brain ventricular transport." *Fluids and Barriers of the CNS* 22.1 (2025): 82). With that model, the

cardiac driving forces had a negligible effect on the LCS compared to the motile cilia forces. Because the cardiac forces in that zebrafish model were modeled by “opening up” the anterior and posterior ventricles using traction boundary conditions for the fluid velocity and pressure, and the fact that it is unclear whether the embryonic zebrafish ventricles at 2 days post-fertilization actually are an “open system”, we wanted to avoid using the cardiac forces in this model, opting for the cilia-only model. Finally, in Herlyng et al. (FBCNS 2025), the cilia model was validated to a greater extent than the cardiac model by comparing with in vivo particle tracking data.
2. Although both cardiac-related forces and muscle contractions may play a role in establishing the LCS, this does not necessarily invalidate the findings that cilia do play a role in establishing the LCS, and that cilia contribute to compartmentalization.

Actions: We have toned down our claims. Specifically, we have renamed some of the result subsection titles to be more clear that our results apply to the model scenario considered, i.e. we have tried to be less general in our statements. We have also added notes in the Discussion that in vivo CSF flow in embryonic zebrafish brain ventricles is likely impacted by cardiovascular-related pulsations and muscle contractions, in addition to the cilia that our model includes.

2. The human model does not take into account modulation of CSF flow by respiratory and cerebrovascular activity. The authors justify this in Section 4.5 with cardiovascular pulsation being the dominant driver, citing Söderström et al. 2025. However, this picture is not clear. Depending on the depth of breathing, respiration can dominate (e.g., Dreha-Kulaczewski et al. 2015), while Wang et al. 2022, NeuroImage 258, 119362, report that cerebrovascular activity produces a larger displaced CSF volume than either cardiac or respiratory action.

Therefore, the flow fields presented in the manuscript constitute a hypothetical case (or, a specific virtual experimental case) rather than in vivo human biology. For the purpose of showcasing FTLE/LCS analysis, this is perfectly fine. However, biological interpretation should be avoided without explicitly stating that the LCS shown reflect the cardiac-driven flow regime in isolation, not the full in vivo flow. Statements about CSF transport, mixing or developmental function should be softened substantially. As with the zebrafish model, a sensitivity analysis could be very valuable.

Comment: We thank the Reviewer for questioning this important point. We acknowledge that the LCS can be influenced by other in vivo factors not considered. We want to first clarify that to a certain degree both respiration and cerebrovascular activity are incorporated in our model. This is because we use experimental brain tissue displacement data as an input for the model; the displacements in those data are from in vivo imaging, and incorporate any mechanism that possibly deforms the brain (except body motion, since the imaged subjects were at rest). However, the displacement data used in the simulations were modeled as being periodic with a period of 1 second (a cardiac cycle), thus in some sense we neglect longer timescale harmonics that may be associated with respiratory or cerebrovascular activity dynamics. Nonetheless, those driving mechanisms are present to some degree in the displacement data in terms of displacement magnitudes.

Actions: We have renamed some of the Results and Discussion subsection titles to soften our statements. We have added to the limitations section info about the displacement data considerations, where we are explicit about the lack of directly including respiration as a mechanism.

3. The finding that inertia is crucial is based on comparably high velocities. The Navier-Stokes / unsteady Stokes comparison (Section 3.6) is conducted at a peak aqueductal velocity of 6.8 cm/s. The authors acknowledge in the Discussion that several other CSF studies report aqueductal velocities a factor of ten lower, and state that "investigating the effects on the discrepancy in the advective transport between the Stokes and Navier-Stokes solutions" is deferred to future work.

The authors argue that a ten-fold velocity reduction leaves Re on the order of 10, so inertia still dominates viscous forces. However, Re is not the only relevant parameter. As the authors note, the unsteady Stokes equation keeps the local acceleration and only loses the convective term, and NS and Stokes models thus only differ through convective inertia. The importance of convective inertia relative to local acceleration scales with the inverse of the Strouhal number, and thus linearly with velocity (at a given driving frequency and geometry). Therefore, whether the qualitative FTLE finding still applies at mm/s velocities cannot be demonstrated with Re alone. Consequently, one of the main findings of this work, that inertia is crucial for transport, may be limited to certain physiological conditions.

Again, the analysis itself is interesting, but the biological (in this case biophysical) claims should be qualified. I would suggest that the authors carry out this comparison in terms of the Strouhal number and repeat the NS versus unsteady Stokes FTLE comparison at peak velocities spanning the full range reported in the literature. Even if the inertia conclusion does not pass this test, it does not mean that, in vivo, inertia is irrelevant, because further driving forces also contribute to CSF flow.

Comment/Action: We thank the reviewer for commenting on this. We have revised our discussion of these findings to not only discussing the effects of the Reynolds number on these results, but also discussing the importance of considering the Strouhal number.

4. The aqueductal stroke volume is used both as a calibration target and as a validation outcome. Section 2.5.1 describes that the wall displacement magnitudes, obtained from experiments, were reduced to avoid "physically unrealistic large amounts of CSF flow". To this end, the rescaling was calibrated against experimental aqueductal stroke volume measurements. The reported stroke volume of 46.1 μL (Section 3.1) is therefore a tuned outcome rather than a prediction. This should be stated prominently in the Methods (with the calibration target value and the scaling factors explicitly given). Because the stroke volume was used to calibrate the displacement magnitude, it cannot serve as an independent validation of the model, and should not be reported as such. It would be more informative to validate against quantities that were not used for calibration.

Comment: We thank the reviewer for pointing this out. We did not intend to present the aqueductal stroke volume as a validation outcome or target, since, as the Reviewer pointed out, this stroke volume was used as a calibration target for the model.

Actions: We have added a reminder for the reader in Results, Sec. 3.1that the aqueductal stroke volume was used as a calibration target and is therefore a tuned outcome. We have stated explicitly in Methods, Sec. 2.5.1 that the aqueductal stroke volume is used as a calibration target, specified the calibration target range, and specified the scaling factors used to adjust the experimental displacement data.

5. The model treats all openings of the fourth ventricle as traction-free open boundaries. This decouples the ventricular system from the subarachnoid space, which is crucial for the physiological partitioning of the outflow. The reported partitioning (lateral 45 μL, median 8.6 μL, Section 3.1) therefore reflects the relative cross-sectional areas and positions of the three openings in the mesh rather than a relevant physiological metric. The authors do address this limitation in the Discussion, but the partitioning values are reported in the Results without a caveat. A sensitivity analysis could help clarify how much of the partitioning stems from the geometry and how much from downstream coupling.

Comment: Thanks for this keen observation. As the Reviewer points out, we acknowledge this limitation in the discussion. During our work, we considered using a type of resistance boundary condition rather than a traction-free boundary condition, to better reflect the coupling to downstream CSF compartments like the subarachnoid space. We thought of applying a model for the downstream CSF compartments e.g. like the one applied in Segers, Patrick, et al. "Systemic and pulmonary hemodynamics assessed with a lumped-parameter heart-arterial interaction model." *Journal of engineering mathematics* 47.3 (2003): 185-199. Better handling of these boundary conditions, or even better including the downstream flow compartments in the model geometry, would be a reasonable refinement in future models.

Action: We have added a remark in Results, Sec. 3.1 stating that the distribution of CSF flow reflects the mesh geometry properties. We have also added a more extensive remark on this in the Limitations subsection of the Discussion.

6. The authors frame the analysis repeatedly in terms of transport and mixing, but no solute transport equation is solved. FTLE fields do, by themselves, not determine clearance or solute residence time, particularly because molecular diffusion is not negligible in all parts of the ventricular space (both human and especially zebrafish). Therefore, the discussion should use advective transport barriers in the strict kinematic sense, rather than in terms of net solute transport, mixing, residence time etc. (quantities that depend on diffusion and cannot be inferred only from FTLE fields). Alternatively, the authors could calculate advection-diffusion of passive tracers of sizes relevant to the current discussion in the literature on brain clearance.

Comment: It is true that FTLE fields do not, by themselves, determine clearance or solute residence times, and that diffusion would also influence these. Therefore, we did our best to refer to advective transport and advective transport times in the manuscript. Diffusion would indeed complicate the transport process more than what is observed when modeling transport as purely advective. We do however think that the purely advective case is worth studying, and that this gives insights on advective transport regimes, which are relevant for certain regions of the brain and certain physiological scenarios.This can even be the case in the zebrafish brain: The same flow model we used in our manuscript was used to study solute transport by an advection-diffusion equation in zebrafish in Herlyng et al. (FBCNS, 2025), where for some physiologically relevant solutes, transport was dominated by advection.
It would indeed be interesting to predict transport of passive tracers by an advection-diffusion equation, but since we regard the current manuscript as already being quite extensive, we chose not to consider such a transport equation in the present work.

Action: To acknowledge the limitation raised by the reviewer, we have added the following remark in the Discussion, end of first paragraph of Section 4.1, to emphasize that diffusion indeed also plays a role in brain ventricular transport:

“Also, as a general point in regards to mixing, the LCS, as computed herein, only act as advective transport barriers; diffusion would also play a role in terms of net solute transport, mixing and residence times in the brain ventricles.” We have also tried to be more explicit in referring to the current results as characteristics of a purely advective transport regime.

Minor points:
Line 35: In humans, CSF does not necessarily flow into the spinal cord, as the central canal is not generally patent. Spinal subarachnoid space would be more appropriate here.

Comment/Action: To make our sentence more generally appropriate, we have rephrased the sentence to “CSF is produced by choroid plexus in the brain ventricles, from where the fluid flows into the subarachnoid space”.

L39: While blockage is one possible cause of hydrocephalus, this condition is more generally a result of an imbalance of CSF production and efflux.

Comment: Here we have also rephrased to make our sentence more generally correct: “Meanwhile, several pathologies afflicting the brain are related to disruption of CSF flow, such as hydrocephalus, where ventricular enlargement results from an imbalance of CSF production and efflux, e.g. by blockage of CSF flow.”

L87: The anatomical similarity between the embryonic zebrafish and adult human ventricular systems is overstated. Zebrafish embryos have unpaired ventricles, compared to the paired lateral ventricles in humans. They also lack frontal, temporal, occipital horns.

Comment/Action: We acknowledge that we overstated the anatomical similarity here. We have therefore rephrased to “... embryonic zebrafish, an animal whose ventricular system, despite being simpler, shares particular conserved developmental and functional features with the human brain ventricles.”

L169: Damping factors change from ζM, ζK to ηM, ηK
L173: "eiqenfrequencies" -> eigenfrequencies

Action: Thank you. Both these typos have been corrected.

L287: Can slip-layer depth be equated to cilium length?

Comment: In our model, we assume that the epithelial cell layer from which the cilia protrude constitutes a noslip surface, in which case the answer is yes. This is however a model assumption we have not validated, which is why we have a rather lengthy discussion in the model limitations section discussing the slip boundary condition vs. a noslip boundary condition, as well as the lack of good-quality validation data for ciliary forces.

L610: Is the vortex ring observed in 2D slices (in which case 'recirculation zone' would be more appropriate) or in 3D? If it is the latter, please show 3D rendering.

Comment: We computed FTLE fields in 3D, and we observe the vortex ring as a 3D flow feature. However, presenting the full 3D field in the manuscript is difficult. However we have clarified in the manuscript (as noted below) that the computations were performed in 3D, and 2D sections of the 3D fields are shown.

Figures 6, 7, 8, 10: Please state whether ridge extraction was performed in 3D or on 2D slices.

Comment/Action: We have specified in the figure captions that the FTLE field computations were performed in 3D, and that the 2D slices presented are extracted from the 3D computations.

Additional actions taken by the Authors

We have revised our Appendix, as we found that parts of it could benefit from improved quality.